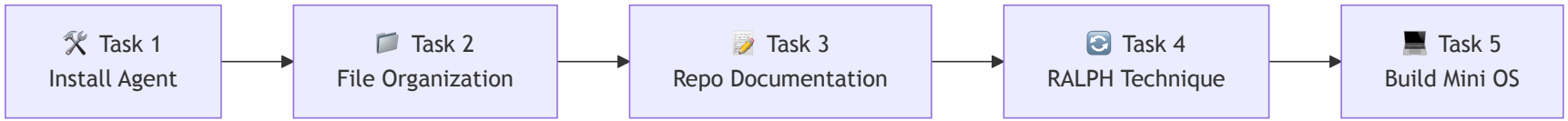
Operating Systems Lab

Week 1 – Coding Agents

Korea University Sejong Campus, Department of Computer Science

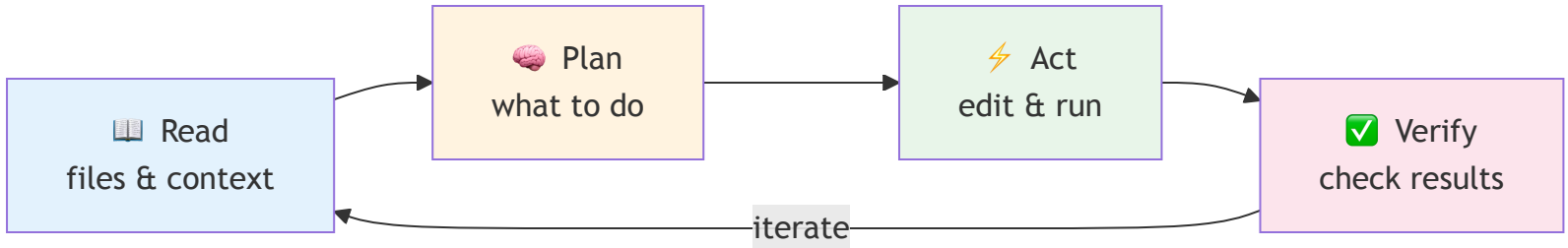
Lab Overview

- **Goal**: Get comfortable using AI-powered coding agents
- **Duration**: ~50 minutes
- **Submission**: None – exploration lab
- **5 Tasks**: Install → File Org → Repo Docs → RALPH → Mini OS



What Are Coding Agents?

AI-powered CLI tools that understand and generate code **in context**



- Can read your file system, run commands, and make edits autonomously
- Useful for: scaffolding, refactoring, documentation, debugging

> **Example**: *"Create a Makefile for a C project with debug and release targets"* Agent: reads directory → writes Makefile → confirms build succeeds

Available Agents

Gemini CLI

Free

Google's agent

Good default choice

Claude Code

Paid

Anthropic's agent

Strong multi-file reasoning

Codex CLI

Paid

OpenAI's agent

Open-source CLI

> **Recommendation**: Install **Gemini CLI** if you have no preference – it's free and easy to set up.

Task 1 & 2: Install & File Organization

Task 1 – Install a Coding Agent

1. Follow the official install guide
2. Verify: `gemini --version`
3. Authenticate with your account

Task 2 – File Organization

Point the agent at a messy directory:

```
"Organize the files in ./downloads
 into subfolders by type and date"
```

Review – did the agent make sensible decisions?

Task 3 & 4: Repo Docs & RALPH

Task 3 – GitHub Repo Documentation

- Open an existing GitHub repository
- Ask the agent to generate a `README.md`
- Check: does it accurately describe the project?

Task 4 – RALPH Technique

Request → **A**nalyze → **L**ist issues → **P**rompt again → **H**armonize

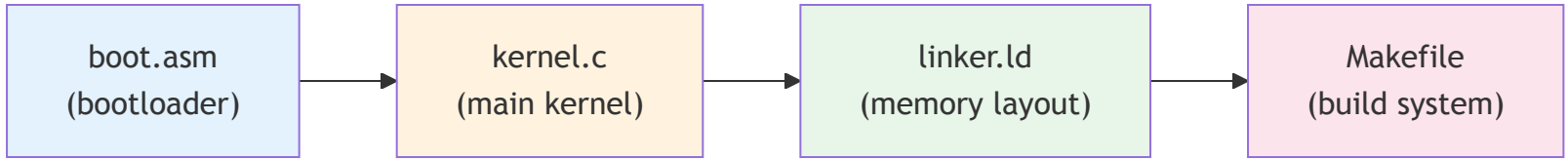
1. Make an initial request to the agent
2. Critically review the output
3. List specific problems or gaps
4. Re-prompt with targeted corrections
5. Repeat until satisfactory

Practice this loop on the README from Task 3.

Task 5: Build a Mini OS

Ask your coding agent to scaffold a minimal OS kernel in C:

```
"Create a minimal x86 OS kernel in C that boots with GRUB,
 prints 'Hello, OS!' to the screen, and halts.
 Include a Makefile and linker script."
```



- Observe how the agent breaks the problem into files
- You do **not** need to successfully boot it – the **process** matters

Summary & Next Steps

What we practiced today

Task	Skill
Install agent	Tool setup & authentication
File organization	Delegating tasks to AI
Repo documentation	Evaluating AI-generated output
RALPH technique	Iterative refinement loop
Mini OS	Tackling complex systems projects with AI

Coming up – Week 2 Lab: Process system calls (`fork` , `exec` , `wait` , `pipe`)

> Coding agents are tools – understanding what they produce is still **your** responsibility.